AI-Based Student Performance Prediction and Academic Risk Analysis System

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***Abstract*—The prediction of student performance is an important application of Machine Learning (ML) in education, particularly for the early identification of academically at-risk students. This paper proposes an AI-Based Student Performance Prediction and Academic Risk Analysis System to classify students into safe and at-risk groups using academic, behavioral, and demographic data. The system was evaluated on a dataset of 5,000 students using structured preprocessing, feature engineering, categorical encoding, standardization, and stratified splitting. Logistic Regression, Random Forest, XGBoost, CatBoost, Stacking, and Soft Voting models were developed and compared. The final system, configured as a Soft Voting Ensemble evaluated at an optimized cross-validated threshold of 0.462, achieved 80.0% accuracy, 72.0% precision, 83.0% recall, 77.0% F1-score, and 0.886 ROC-AUC. Odds Ratio and TreeSHAP identified Project Score, Academic Average, and Midterm Score as protective factors, while Stress Level was the most significant risk factor. The system supports explainable early risk detection and timely academic intervention.**

***Index Terms*—Student Performance Prediction, Machine Learning, Educational Data Mining, Learning Analytics, Academic Risk Analysis, Early Warning System.**

I. Introduction

Personal and professional development is very important through education. It assists individuals to develop confidence and skills and improve their social and economic conditions. A student's academic performance is an important factor in determining progress in higher education. It also impacts future job prospects, university reputation, and university rankings. However, universities face several challenges in student performance monitoring, delivering quality education, and predicting future needs [1, 2].

Although many universities have invested in student support and retention programs, they still face problems with student dropout and early attrition. Student dropouts affect not only individual students but also the university's operational efficiency, financial stability, and reputation [3, 2]. Traditionally, universities have implemented academic support and mentoring programs to reduce dropouts. However, these methods tend to be reactive rather than proactive. Intervention often begins when warning signs appear, such as poor exam performance, low attendance, or missed assignments. Furthermore, traditional Student Management Systems (SMS) typically rely on data collected after one or more semesters, including GPA, failed courses, and enrollment history. Therefore, universities are often unable to detect struggling students early in their educational process [3, 4].

Recent advancements in machine learning and Learning Analytics (LA) provide new opportunities to address this problem. The widespread use of Learning Management Systems (LMS), Virtual Learning Environments (VLE), and Student Information Systems (SIS) generates large amounts of data about students' learning activities and behaviours. This data can be used to develop Early Warning Systems (EWS) using machine learning to identify students at risk of academic failure at an early stage [5, 2].

However, developing an effective EWS remains challenging. Selecting appropriate machine learning models is a significant concern. Traditional methods such as Logistic Regression, Naïve Bayes, and Decision Trees are easy to understand and explain, while advanced models such as Random Forest, Gradient Boosting, XGBoost, and Artificial Neural Networks

(ANN) can handle complex patterns in data. These advanced models are often considered "black boxes" because their decisions are difficult to interpret. Therefore, Explainable Artificial Intelligence (XAI) approaches such as SHAP and LIME are important in educational prediction systems [1, 6, 2]. Another challenge is educational data quality. Real-world datasets are often imbalanced, with more successful students than students at risk. Moreover, academic records, socioeconomic status, and e-behavior can influence student performance. Integrating behavioral data such as login frequency and resource usage with demographic and academic information can enhance prediction effectiveness [4, 1, 6].

To overcome these challenges, this paper presents an AI-Based Student Performance Prediction and Academic Risk Analysis System structured around the CRISP-DM framework. The primary contributions of this work are fourfold:

1) **Leakage-Free Pre-Outcome Modeling:** We formulate an early-to-mid semester prediction pipeline on $N = 5{,}000$ student records by discarding post-outcome academic indicators and deriving domain-specific wellness metrics (Burnout_Risk and Study_to_Sleep_Ratio) that integrate physiological recovery with academic effort.
2) **Multi-Family Ensemble Architecture:** We integrate linear models (Logistic Regression) with diverse tree-based ensembles (Random Forest, XGBoost, CatBoost) via a Soft Voting meta-combination, capturing linear trends alongside non-linear multi-feature interactions.
3) **Strict Cross-Validated Threshold Optimization:** To prioritize sensitivity without data leakage, we calibrate the decision cutoff ($\tau^* = 0.462$) exclusively on 5-fold cross-validation out-of-fold training probabilities, evaluating the calibrated pipeline once on an untouched hold-out test set ($N_{\text{test}} = 1{,}000$).
4) **Dual-Layer Explainability:** We bridge global parametric interpretations via Logistic Regression Odds Ratios (OR) with local non-linear feature attributions using TreeSHAP, resolving apparent discrepancies between raw marginal statistics and multivariate interaction effects.

The paper proceeds as follows: Section II reviews existing literature on student performance tracking. Section III describes the methodology, Section IV reports and discusses the findings, and Section V concludes with a summary and future directions.

II. Related Works

Student performance prediction is an important area in Educational Data Mining (EDM) and Learning Analytics (LA). Machine learning is used to analyze students' academic and behavioral data to identify at-risk students and improve academic performance [2].

Ahmed [7] applied SVM, Decision Tree, Naïve Bayes, and KNN with K-means clustering on data from 32,005 students. SVM achieved the highest accuracy of 96.03%, but the pipeline was limited by single-institution data and parameter sensitivity.

Alkhatib et al.[1] found that ensemble models such as Random Forest and Gradient Boosting performed better than linear models, while simpler models were easier to interpret.

Carballo-Mendívil et al. [3] developed a CRISP-DM-based framework using Logistic Regression, Random Forest, LightGBM, and XGBoost on pre-enrollment admissions data. Their calibrated XGBoost model achieved 88% recall with a decision threshold of $\tau = 0.3380$.

Ouatik et al. [5] integrated big-data computing (Hadoop MapReduce and HDFS) with machine learning, showing that combining Sequential Minimal Optimization (SMO) feature selection with SVM achieved 87.32% accuracy while accelerating runtime.

Albreiki et al. [4] presented a systematic literature review of studies between 2009 and 2021. Their review identified Decision Trees, Naïve Bayes, and SVM as common algorithms, while noting challenges including class imbalance and sparse temporal feature tracking.

Wang [6] proposed a hybrid stacking model combining Decision Trees, Random Forest, SVM, and ANN, achieving over 96% accuracy across multiple benchmarks (peaking at 98.8% on the UCI dataset).

Janahi and Obeidat [2] reviewed institutional case studies and found that early prediction can help students receive support earlier. They also highlighted privacy, ethical data use, and reducing black-box issues

Sheikh et al. [8] evaluated Gradient Boosting, Random Forest, Agentic AI, and LLMs using LMS logs from 320 undergraduate students. Gradient Boosting achieved 0.90 accuracy and 0.94 AUC-ROC, validating the utility of engagement metrics for risk detection.

III. Methodology

As illustrated in Fig. 1, the proposed academic early warning system is structured into three main phases: i) dataset preprocessing and domain-specific feature engineering, ii) machine learning model development and ensembling, and iii) model evaluation, threshold optimization, and explainability analysis.

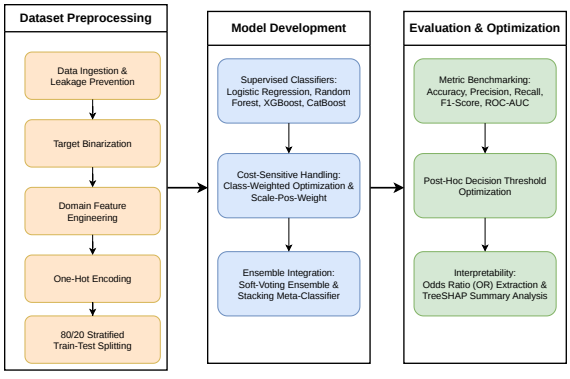


Fig. 1: Workflow of the proposed early academic risk prediction framework.

A. Dataset Preprocessing & Feature Engineering

The experimental evaluation utilizes the Students Performance Dataset [9]. The pipeline enforces strict data integrity rules prior to model exposure:

TABLE I: Summary of Related Works

Ref.	Year	Dataset	Method	Result	Limitations
[7]	2024	Wollo University ($N = 32{,}005$)	K-Means, SVM, DT, NB, KNN with Grid Search	SVM achieved 96.03% accuracy, DT reached 93.41%	Single institution; sensitive to parameter tuning
[4]	2021	Systematic Review (78 papers)	Review of DT, RF, SVM, ANN, KNN, LR	DT, NB, and SVM most prevalent; reported accuracies 60%–90%+.	Limited training sets, class imbalance, sparse temporal features
[1]	2025	Comparative Survey (2020–2024)	Comparative analysis of LR, DT, RF, SVM, ANN	Ensembles (RF, GBM) yield highest accuracy and F_1-scores	High-accuracy models exhibit opacity; lacks emotional/lifestyle features
[3]	2025	Pre-enrollment Data ($N = 38{,}932$)	CRISP-DM, LR, RF, LightGBM, XGBoost	Calibrated XGBoost achieved 88% recall ($F_1 = 0.69$, $\tau = 0.3380$)	Dependent on self-reported survey data; binary dropout target
[5]	2022	VLE & Academic Logs (Morocco)	Hadoop MapReduce, HDFS, SMO, SVM, KNN, C4.5	SVM with SMO selection achieved 87.32% accuracy	High infrastructure dependency; restricted to single institution
[8]	2026	Digital LMS Logs ($N \approx 320$)	Gradient Boosting, RF, Agentic AI, LLMs	Gradient Boosting achieved 0.90 accuracy, 0.86 recall, 0.94 AUC	Small-scale cohort; lack of physiological and affective variables
[6]	2025	UCI, OULAD, NELS:88	Hybrid Stacking (DT, RF, SVM, ANN)	Stacking achieved 98.8% (UCI), 96.9% (OULAD), 97.5% (NELS)	Sensitive to class imbalance; ethical and data governance concerns

- **Data Leakage Mitigation:** Non-predictive personal identifiers (`Student_ID`, `First_Name`, `Last_Name`, `Email`) and post-outcome variables (`Grade`, `Final_Score`, `Total_Score`) were removed from the training partition to ensure the framework operates strictly on early-to-mid-semester indicators.
- **Target Formulation:** Academic risk is formulated as a binary classification task defined by end-of-term academic standing:

$$y = \begin{cases} 1, & \text{(at risk), if Grade} \in \{\text{D}, \text{F}\} \\ 0, & \text{(safe), if Grade} \in \{\text{A}, \text{B}, \text{C}\} \end{cases} \tag{1}$$

- **Domain Interaction Features:** Four composite features were engineered to capture student consistency and lifestyle balance:
  - *Academic Performance Indices ($\mu_{acad}, \sigma_{acad}$):* Mean and standard deviation computed across continuous coursework components (Assignments_Avg, Quizzes_Avg, Participation_Score, Projects_Score).
  - *Midterm-to-Coursework Gap (Δ_{exam}):* Discrepancy between midterm examination scores and continuous coursework averages:

$$\Delta_{\text{exam}} = \text{Midterm_Score} - \mu_{\text{acad}} \tag{2}$$

  - *Effort-to-Recovery & Burnout Indicators ($R_{effort}, I_{burnout}$):* Measures student workload sustainability by evaluating dedicated study time against essential sleep and recovery:

$$\text{Burnout_Risk} = \frac{\text{Stress_Level (1–10)}}{\text{Sleep_Hours_Per_Night} + \epsilon} \tag{3}$$

$$\text{Study_to_Sleep_Ratio} = \frac{\text{Study_Hours_per_Week}}{\text{Sleep_Hours_Per_Night} + \epsilon} \tag{4}$$

  - *Attendance Risk Flag (I_{att}):* Binary indicator flagging severe absenteeism:

$$I_{\text{att}} = \mathbb{I}(\text{Attendance_Rate} < \tau_{\text{att}}) \tag{5}$$

- **Transformation Pipeline:** Categorical variables were encoded using dense One-Hot Encoding (`handle_unknown='ignore'`), while numerical variables were normalized via Z-score standardization:

$$z = \frac{x - \mu}{\sigma} \tag{6}$$

The dataset was partitioned into an $80:20$ train-test split stratified on the target label y.

B. Model Development

To capture both linear boundaries and higher-order feature interactions, five supervised learning architectures and ensemble strategies were developed:

1) **Tuned Logistic Regression (LR):** A regularized generalized linear model estimating class posterior probabilities via the logit link function:

$$P(y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \boldsymbol{\beta}^T \mathbf{x})}} \tag{7}$$

Cost-sensitive class weighting was applied to mitigate penalty bias:

$$w_j = \frac{N}{2 \cdot N_j} \tag{8}$$

2) **Random Forest (RF):** A bootstrap-aggregated ensemble of decorrelated decision trees using random subspace feature selection:

$$\hat{y} = \text{mode}\left\{h_b(\mathbf{x})\right\}_{b=1}^{B} \tag{9}$$

3) **Extreme Gradient Boosting (XGBoost):** A scalable tree-boosting system minimizing a regularized objective via second-order Taylor approximations:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} \left[g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i) \right] + \Omega(f_t) \tag{10}$$

where g_i and h_i denote first- and second-order gradient statistics, and `scale_pos_weight` accounts for minority class imbalance.

4) **Categorical Boosting (CatBoost):** An implementation of gradient boosted trees leveraging oblivious decision trees and ordered target statistics:

$$\hat{y}^{(t)} = \hat{y}^{(t-1)} + \eta f_t(\mathbf{x}) \tag{11}$$

5) **Soft-Voting Ensemble:** Combines probability distributions across linear and tree-based architectures:

$$\hat{P}(y = 1 \mid \mathbf{x}) = \frac{1}{M} \sum_{m=1}^{M} P_m(y = 1 \mid \mathbf{x}) \tag{12}$$

C. Performance Evaluation

The overall performance was evaluated using five widely adopted classification metrics: Accuracy, Precision, Recall, F_1-score, and ROC-AUC:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{13}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{14}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{15}$$

$$F_1\text{-Score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{16}$$

$$\text{ROC-AUC} = \int_0^1 \text{TPR}\left(\text{FPR}^{-1}(t)\right)\, dt \tag{17}$$

To minimize missed struggling students (False Negatives), post-hoc decision threshold optimization was performed across the precision-recall curve:

$$\tau^* = \arg\max_{\tau \in (0,1)} F_1(\tau) \tag{18}$$

To prevent data leakage, all hyperparameters and decision thresholds were calibrated exclusively on the training partition ($N_{\text{train}} = 4{,}000$) using 5-fold stratified cross-validation. Sweeping decision boundaries across out-of-fold cross-validation probability distributions identified the optimal threshold at $\tau^* = 0.462$, which maximized harmonic balance (F_1^{CV}). The Final System is formally defined as the Soft Voting Ensemble evaluated at this frozen cross-validated cutoff $\tau^* = 0.462$ once on the independent hold-out test set ($N_{\text{test}} = 1{,}000$). Model interpretability was established through odds ratio ($\text{OR} = e^{\beta}$) extraction and TreeSHAP feature attributions.

D. Summary Table

TABLE II: Dataset and Experimental Setup Summary

Item	Description
Dataset	Students Performance Dataset from Kaggle
Samples (N)	5,000
Engineered Features	20+ (Demographic, Coursework, Wellness Ratios, Interaction Flags)
Target	Academic Risk ($y \in \{0, 1\}$)
Class Balance	Safe: 3,035 (60.7%), At-Risk: 1,965 (39.3%)
Partitioning	Stratified Train/Test Split (80 : 20)
Preprocessing	Leakage dropping, One-Hot Encoding, `StandardScaler`
Imbalance Handling	Cost-sensitive class weighting (`balanced`, `scale_pos_weight`)
Optimization	Decision threshold tuning ($\tau^* = 0.462$)
Validation	Hold-out test set ($N_{\text{test}} = 1{,}000$) and 5-Fold Stratified CV

IV. Results and Discussion

A. Dataset Distribution and Statistical Profiling

The experimental benchmark was evaluated on the Students Performance Dataset ($N = 5{,}000$). The target variable reflects a realistic academic distribution: 3,035 students (60.7%) belong to the safe cohort ($y = 0$), while 1,965 students (39.3%) fall into the at-risk category ($y = 1$), yielding a class ratio of approximately $1.54 : 1$.

TABLE III: Descriptive Statistics Across Feature Dimensions and Class Cohorts

Feature	Total Mean ± SD	Safe ($y = 0$) Mean ± SD	At-Risk ($y = 1$) Mean ± SD
Midterm_Score	70.70 ± 17.44	74.34 ± 16.88	65.42 ± 16.87
Assignments_Avg	74.96 ± 14.40	78.12 ± 14.13	70.36 ± 13.53
Quizzes_Avg	74.84 ± 14.42	76.71 ± 14.20	72.12 ± 14.31
Projects_Score	74.78 ± 14.54	80.49 ± 13.31	66.49 ± 12.05
Attendance (%)	75.36 ± 14.39	75.25 ± 14.41	75.52 ± 14.37
Study_Hours_per_Week	17.52 ± 7.19	17.56 ± 7.20	17.46 ± 7.18
Stress_Level (1–10)	5.51 ± 2.89	5.56 ± 2.87	5.43 ± 2.91
Sleep_Hours_per_Night	6.51 ± 1.45	6.53 ± 1.44	6.50 ± 1.45

As shown in Table III, clear distributional shifts emerge between classes. Safe students demonstrate higher continuous academic averages (e.g., `Projects_Score` of 80.49 ± 13.31 vs. 66.49 ± 12.05 and `Midterm_Score` of 74.34 ± 16.88 vs. 65.42 ± 16.87). Demographic factors remain evenly split across both cohorts, indicating that predictive utility stems primarily from continuous academic indicators and dynamic engagement patterns rather than static background traits.

B. Experimental Results and Model Comparison

Model stability was first benchmarked across 5-fold stratified cross-validation on the training set ($N_{\text{train}} = 4{,}000$) prior to hold-out evaluation. Cross-validation results confirmed robust generalization: Tuned Logistic Regression achieved an average CV accuracy of 0.801 ± 0.007 and ROC-AUC of 0.885 ± 0.006; Tuned CatBoost achieved 0.797 ± 0.009 accuracy and 0.880 ± 0.007 ROC-AUC; and the Soft Voting ensemble achieved 0.800 ± 0.006 accuracy and 0.884 ± 0.008 ROC-AUC.

Across the hold-out test set ($N_{\text{test}} = 1{,}000$), baseline classifiers operating at $\tau = 0.50$ exhibited high false-negative rates, with Decision Tree performing lowest (0.697 accuracy, 0.639 F_1-score; Table III(a)). Tuned Logistic Regression established the strongest standalone baseline at the default threshold $\tau = 0.50$ (0.802 accuracy, 0.768 F_1-score, 0.887 ROC-AUC; Table III(b)). However, Tuned LR's default sensitivity leaves nearly 20% of vulnerable students undetected (0.801 recall). Tree-based models (Tuned CatBoost, XGBoost, and Random Forest) attained comparable ranking metrics (ROC-AUC 0.878–0.884).

The Soft Voting Ensemble aggregates posterior class distributions across linear and tree architectures, yielding well-calibrated continuous risk scores. Consequently, shifting the operational cutoff to the cross-validated optimum ($\tau^* = 0.462$) enables the Final System to achieve superior at-risk detection sensitivity (0.830 recall, 0.770 F_1-score, 0.800 accuracy, and 0.886 ROC-AUC) compared to Tuned LR evaluated at the same cutoff (0.801 recall, 0.768 F_1-score). In academic early warning systems, maximizing recall on struggling students while keeping false alarms low justifies adopting the ensemble over a standalone linear classifier.

As illustrated in Fig. 2a, all trained classifiers significantly outperformed the random baseline (AUC = 0.500). Tuned Logistic Regression (AUC = 0.887) and the Soft Voting Classifier (AUC = 0.886) achieved the highest area under

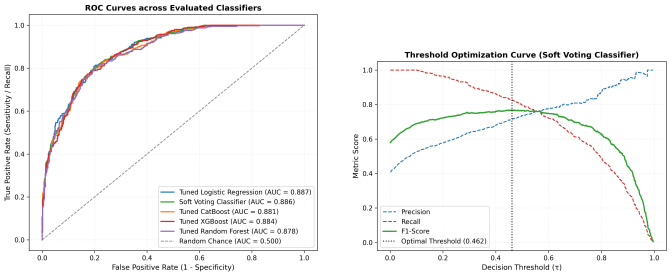
TABLE IV: Benchmark Results Before vs. After Optimization (Hold-Out Test Set)

(a) Baseline Performance ***($\tau = 0.50$, Unweighted)***

Model	Accuracy	Precision	Recall	F_1-Score	ROC-AUC
Decision Tree	0.697	0.622	0.657	0.639	0.691
XGBoost	0.789	0.751	0.723	0.737	0.874
Random Forest	0.796	0.770	0.713	0.740	0.868
CatBoost	0.799	0.770	0.723	0.746	0.881
Logistic Regression	0.803	0.773	0.733	0.752	0.888

(b) Optimized Pipeline (Engineered, Weights, ***$\tau^* = 0.462$)***

Model	Accuracy	Precision	Recall	F_1-Score	ROC-AUC
Tuned XGBoost	0.792	0.724	0.792	0.756	0.884
Tuned Random Forest	0.798	0.738	0.782	0.760	0.878
Tuned CatBoost	0.799	0.737	0.789	0.762	0.881
Stacking Classifier	0.798	0.728	0.806	0.765	0.886
Soft Voting Classifier	0.801	0.737	0.797	0.766	0.886
Tuned LR	0.802	0.736	0.801	0.768	0.887
Final System (Soft Voting$\|\tau^* = 0.462$)	0.800	0.720	0.830	0.770	0.886



(a) (b)

Fig. 2: Performance metrics curves: (a) ROC curves comparing evaluated classifiers, (b) Threshold optimization dynamics for the Soft Voting Classifier.

the curve, confirming strong ranking stability for identifying at-risk individuals.

C. Decision Threshold Optimization

Under standard classification settings with a default threshold ($\tau = 0.50$), model sensitivity for the minority positive class ($y = 1$) hovered around 79.7%, leaving approximately one out of every five vulnerable students undetected.

As shown in Fig. 2b, sweeping the decision threshold reveals that shifting the classification cutoff to $\tau^* = 0.462$ achieves the peak F_1-score (0.770), creating a balanced trade-off between sensitivity and precision.

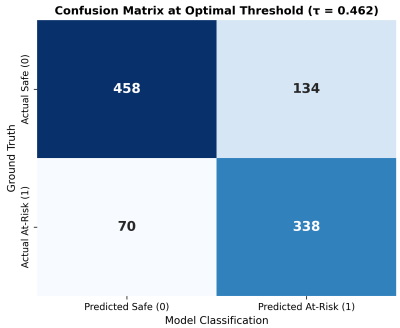


Fig. 3: Confusion matrix for the Soft Voting Classifier evaluated at the calibrated operational threshold ($\tau^* = 0.462$).

Applying this optimized threshold ($\tau^* = 0.462$) yielded the confusion matrix presented in Fig. 3. The system successfully detected 338 out of 408 true at-risk students, boosting recall to 82.8% with a precision of 71.6% (458 true negatives, 134 false alarms, and only 70 false negatives).

D. Feature Attribution and Explainability Analysis

To ensure institutional transparency, model predictions were analyzed through TreeSHAP summary distributions and Logistic Regression Odds Ratios ($\text{OR} = e^{\beta}$).

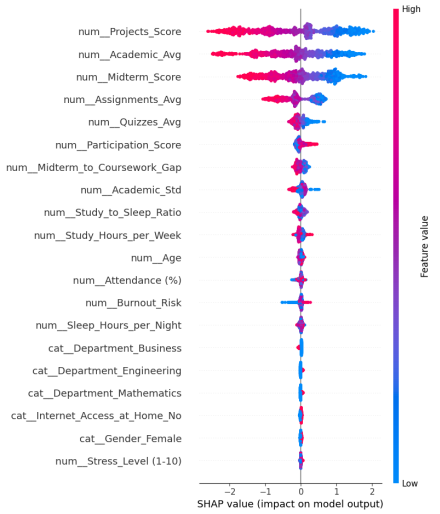


Fig. 4: TreeSHAP summary beeswarm plot illustrating the directional and magnitude impact of student features on the risk prediction output.

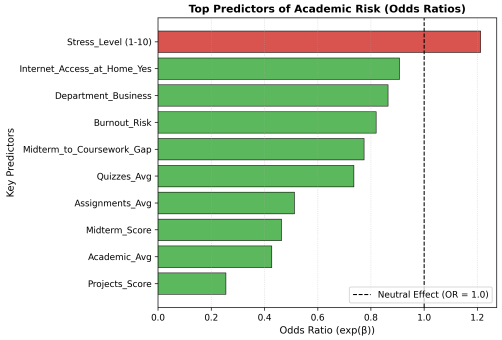


Fig. 5: Top predictors of academic risk ranked by Logistic Regression Odds Ratios ($\text{OR} = e^{\beta}$), where $\text{OR} > 1.0$ indicates risk amplification and $\text{OR} < 1.0$ represents protective influence.

Both explainability frameworks demonstrate strong agreement regarding the primary determinants of academic standing:

- **Primary Protective Factors (OR $<$ 1.0, Negative SHAP Impact):** As visualized in Fig. 4 and Fig. 5, continuous coursework performance metrics—specifically Projects_Score (OR $<$ 1.0), Academic_Avg (OR $<$ 1.0), and Midterm_Score (OR $<$ 1.0)—exerted the strongest

protective effects. Higher marks in these areas correspond to negative SHAP values, significantly reducing the likelihood of academic risk.

- **Multivariate Risk Drivers and Stress Interaction (OR $\approx$ 1.21, Positive SHAP Impact):** In raw univariate descriptive statistics (Table III), the mean stress levels reported appear nearly identical between the safe (5.56 ± 2.87) and at-risk (5.43 ± 2.91) groups. However, in the multivariate logistic model and tree ensembles, Stress Level (1–10) emerges as a significant risk multiplier ($OR \approx 1.21$, positive SHAP attribution). This occurs because stress operates as an interactive suppressor variable: when evaluated alongside sleep duration and study workload through the engineered composite indicators (*Burnout Risk* and *Study-to-Sleep Ratio*), elevated stress substantially increases failure probability among students who already maintain borderline coursework marks.
- **Demographic Context:** Categorical demographic indicators (Internet_Access_at_Home, Department_Business) showed near-neutral odds ratios (OR ≈ 1.0) and condensed, centered SHAP distributions near zero. This confirms that real-time academic assessments and physiological stress metrics are far stronger determinants of course outcomes than static demographic backgrounds.

TABLE V: Performance Benchmark Comparison with Prior Studies

Study / Configuration	Accuracy	Precision	Recall	F_1-Score	ROC-AUC
Carballo-Mendívil et al. (2025) [3]	0.600	0.570	**0.880**	0.695	0.690
Ahmed (2024) [Naïve Bayes Baseline] [7]	0.774	0.819	0.894	0.743	—
Sheikh et al. (2026) [Baseline LR] [8]	0.820	0.760	0.710	0.730	0.860
Albreiki et al. (2021) [EDM Survey Average] [4]	0.60–0.75	—	—	0.60–0.70	0.691
Our Proposed Work (Soft Voting $\mid \tau^* = 0.462$**)**	**0.800**	**0.720**	**0.830**	**0.770**	**0.886**

E. Comparison with Similar Works

Compared to related early warning systems such as Carballo-Mendívil et al. [3], which suffered from lower accuracy (60.0%) and precision (57.0%) despite high recall, our calibrated Soft Voting ensemble achieves a superior balance with 80.0% accuracy, 72.0% precision, 83.0% recall, and an F_1-score of 0.770 . Furthermore, our framework outperforms standard parametric baselines like Ahmed's Naïve Bayes ($F_1 = 0.743$) and Sheikh et al.'s baseline ($F_1 = 0.730$) [7, 8]. By combining multi-family supervised learning with post-hoc threshold tuning ($\tau^* = 0.462$), the system drastically curtails false alarms while providing dual-layer interpretability via TreeSHAP and Logistic Regression Odds Ratios.

V. Conclusion

This study introduced an AI-based early warning system combining supervised machine learning, cost-sensitive reweighting, and post-hoc threshold calibration to predict student performance and identify academic risk. The final system, defined as a Soft Voting ensemble evaluated under a cross-validated threshold ($\tau^* = 0.462$), achieved 80.0% accuracy, 72.0% precision, 83.0% recall, an F_1-score of 0.770, and 0.886 ROC-AUC on an independent hold-out test set ($N_{\text{test}} = 1{,}000$). Continuous coursework evaluations (assignments, quizzes, and project scores) served as primary protective factors, while stress levels in conjunction with sleep and recovery deficits acted as interactive risk amplifiers. These findings confirm that combining multi-model ensembles with calibrated decision boundaries provides reliable, explainable risk detection for educational decision support.

However, the study is limited by the use of a single, moderately imbalanced dataset without longitudinal tracking, which limits tracking academic risk dynamics over subsequent terms. The system also requires formal educational deployment to validate intervention efficacy. Future work will investigate larger cross-institutional cohorts, collect multi-semester longitudinal data, and incorporate adaptive behavioral indicators to enhance predictive reliability, interpretability, and intervention support.

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